

10. When an electrical discharge passes through gaseous hydrogen at low pressure, electromagnetic radiation is emitted.

- (a) Describe the processes within a hydrogen atom that cause electromagnetic radiation to be emitted. [2]

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- (b) If the electromagnetic radiation in part (a) is passed through a spectrometer, several series of converging lines are observed.

- (i) Explain why there are several series of lines. [1]

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- (ii) Explain why the lines within each series converge. [1]

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- (c) The convergence limit of the Lyman series of lines occurs at a wavelength of 91.2 nm.

- (i) State what the limit represents. [1]

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- (ii) Calculate the energy, in kJ mol^{-1} , of the convergence limit. [4]

Energy = kJ mol^{-1}



(d) Hydrogen forms when aluminium reacts with sulfuric acid.



- (i) Calculate the volume of hydrogen, in cm^3 , that would be produced if 0.131 g of aluminium were added to an excess of sulfuric acid at a temperature of 25°C and a pressure of 1 atm. [3]

Volume = cm^3

- (ii) Calculate the volume of hydrogen that would be produced if the same experiment were carried out at 50°C and 1.6 atm. [2]

(If you do not have an answer in part (i), assume that the volume is 200 cm^3 . This is **not** the correct answer.)

Volume = cm^3

